

Statement of Purpose

My journey began when I was three, on the night my grandmother collapsed from a stroke. I grew up watching my parents exhaust themselves caring for her, and I saw the same struggle in hospitals and care facilities among many elderly and disabled people. This was not an exceptional situation; what we called “family duty” was also a structural challenge in aging societies. By middle school, I had already decided where I needed to study to address it: Japan, having entered a super-aged society earlier than most and home to world-leading robotics, offered fertile ground to develop solutions. Right after high school, I left Korea and moved there alone to study robotics, aiming to build assistive robots that could actually make daily care less painful for older adults and people with disabilities. That decision gave me my first proof that this mission is not theoretical for me. Now I am seeking a PhD in the U.S. because, rather than being an engineer who applies existing technologies, I believe core advances in AI and robotics are still needed, and I want to help drive that progress toward assistive robots that families can rely on at home.

In Japan, I completed my undergraduate studies in Mechanical Systems Engineering at Shinshu University, which I chose because Professor Kimitoshi Yamazaki’s work in intelligent robotics for daily life support directly aligned with my goals. Before joining his lab, I immersed myself in diverse courses to prepare for research in robotics. I soon realized that robotics is an interdisciplinary field bridging many branches of science and engineering. Wanting to understand this breadth before choosing my own specialization, I deliberately took on an intensive course load, at one point completing 40 credits in a single semester with special permission from the university. Through coursework in dynamics, control, and robotics, I built a foundation strong enough to model and design robotic systems from the ground up.

Even amid that intensity, I devoted nearly all my remaining time to the university’s robotics club, a small, underfunded team where every member had to master the full engineering pipeline. Despite the harsh constraints of the COVID-19 pandemic, our 14-member team collaboratively planned and built robots for the NHK Japanese national robotics competition, handling everything from CAD design and machining raw steel to wiring, assembly, programming, control, and testing. The competition challenged teams to build robots capable of stacking cylindrical “lagori” blocks into a five-tier tower nearly the height of a person, while also collecting balls and intercepting attacks from the opponent’s robot. I participated across all stages but mainly led software development. Using ROS, I implemented manual teleoperation and developed two autonomous modules: a ball-retrieval system and an opponent-tracking interception system, incorporating a front camera with OpenCV-based movement-detection and a laser range finder (LRF) for basic localization. The ball-retrieval module operated about two seconds slower than manual control, and the interception module had a low success rate; therefore, neither was used in official matches. Nevertheless, the robots impressed the judges, and our team was within the national top five and received a Special Award for innovation and technical excellence.

Through my work on many robots, I realized that adding more sensors often makes systems heavier and more prone to errors. To make robots practical in daily life, they must be lightweight and adaptable rather than dependent on complex hardware. This led me to pursue policy learning without external cameras or force sensors, focusing on plug insertion as one of the tasks that demands both precision and flexibility. After joining Professor Yamazaki’s lab, I explored this idea using NVIDIA Isaac Gym to let a robot learn the entire insertion motion—much like a person fumbling for a keyhole in the dark. In simulation, I used the simulator-provided plug–socket relative pose (the midpoints of the socket holes and plug prongs) as the policy input, and received higher reward as their distance decreased. A 7-DOF Franka arm trained with PPO, with small random offsets added to the plug and socket initial pose in position and orientation at each episode reset, autonomously learned how to approach the socket and gradually refine its motion through trial and error until the plug was successfully inserted. In simulation, I evaluated the learned insertion by replaying the learned trajectory from a near-aligned pre-insertion pose. The learned movement achieved an average 83% success rate, counting an episode as a failure if insertion was not completed within 11 seconds.

During those years, I learned not only the promise of robotics but also its limits. Yet in the NHK competition, as impressive as these robots were, they were also oversized, expensive, and impractical for daily use. In the Yamazaki lab, I observed and even helped present robots designed to put on shirts and pants, including operating a booth at ICRA 2024 that demonstrated a dressing robot. Although the engineering was sophisticated, these projects revealed to me the limits of technical achievement when disconnected from daily human needs. These projects also made clear that practical constraints such as size, cost, safety, and maintenance can limit adoption in daily life. From my experience, technical performance is necessary but not sufficient; real adoption depends on whether robots can act in ways that feel safe, understandable, and emotionally supportive in daily life. I realized we need robots that not only perform tasks like plugging in or dressing a person, but also connect with people emotionally and

build a genuine connection. These experiences convinced me that robotics must not be measured by technical complexity alone, but by how meaningfully it connects with human needs.

In contrast, my experience with support from Sony's toio, which is a cube-shaped robot, opened my eyes to a different possibility. Sony encouraged me to explore creative applications of toio as an example of how toio could inspire playful and educational interaction. I developed a game in which the robots chased and evaded one another in a confined arena. I coded the pursuit logic, attempted crowd-control algorithms to prevent collisions, and created the player-control interface. When children played with these tiny robots, they laughed and even learned from them. Despite their simplicity, these robots had a visible impact on children's joy and education. I realized that the true value of robotics is not confined to physical ability but also lies in its power to evoke emotional connection. From that point, I began to see robotics not only as a set of mechanical systems but as partners capable of shaping human experience.

To pursue this vision, I also needed to ground it in how modern robotics is actually developed. Because robotics research is shared and evaluated largely in English, I wanted to train in an environment where I could learn advanced control and robotics through project-based coursework while also communicating and collaborating in English. That motivation led me to join California State University, Sacramento, as an exchange student. There, I was recognized by several professors and granted special permission to enroll in the most challenging, prerequisite-heavy courses, studying and living just like local students. I focused on control and robotic systems, spending a year in a lab-style course under Assistant Professor Rohollah Moghadam, where I implemented and tuned PID controllers in MATLAB on simple dynamic systems; he later encouraged me to consider pursuing a U.S. PhD. Under Lecturer Neal Frederick Levine, a teammate and I built an automatic-parking robot end-to-end using MATLAB and Arduino-based kits, implementing the motion-control logic ourselves. Concurrently with Assistant Professor Ray Tang, I completed PLC & Motion Fundamentals training in Silicon Valley, learning robot-arm software and system architecture with Beckhoff.

Underlying this interest is a long-term commitment that has guided my path to Japan and then to the United States. It has shaped each major decision I have made, from leaving home after high school to study robotics in a super-aged society, to choosing elder-care work in place of compulsory military service for Korean men, to challenging myself once more through research abroad. Without knowing anyone in Japan or the United States, I repeatedly had to rebuild my environment from scratch, taking ownership of demanding projects in new languages and cultures. Doing so has shown me that I can not only adapt quickly, but also grow into someone who frames problems independently, handles demanding technical work, and learns steadily from failures and outcomes.

At ABC University, I am especially drawn to research that

Ultimately, I aim to help create a society in which it is natural for households to rely on robots as partners in living and caregiving—robots that help older adults and people with disabilities stay in their homes, and that reduce the emotional and practical burden carried by their families. To work toward this vision, in my PhD at ABC University, I want to pursue algorithmic and system-level advances in AI and robotics so assistive robots can be adopted in homes rather than only demonstrated in labs. Building on my experience, I aim to develop assistive autonomy and interaction designs that remain safe, understandable, and psychologically comfortable for users under real constraints such as uncertainty and maintenance. I want to study how robot behavior and interaction should be designed so assistance reduces burden for families instead of adding new friction. ABC's strengths spanning computational robotics for real-world autonomy and human-centered interaction research make it a natural fit to develop and validate this integrated direction.